Introduction to Programming

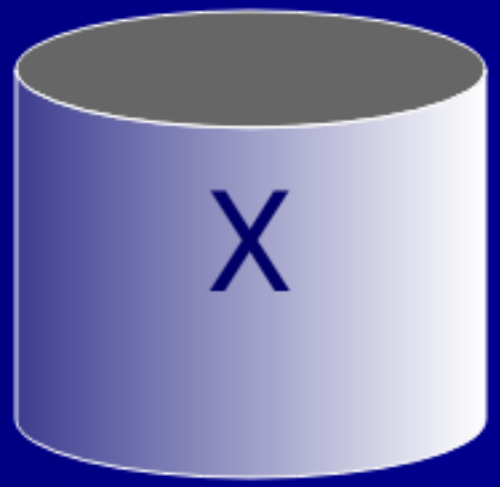
Lesson 3

VU
Virtual University

```
#include <iostream.h>
main ( )
{
 cout << " Welcome to Virtual University ";
}
```

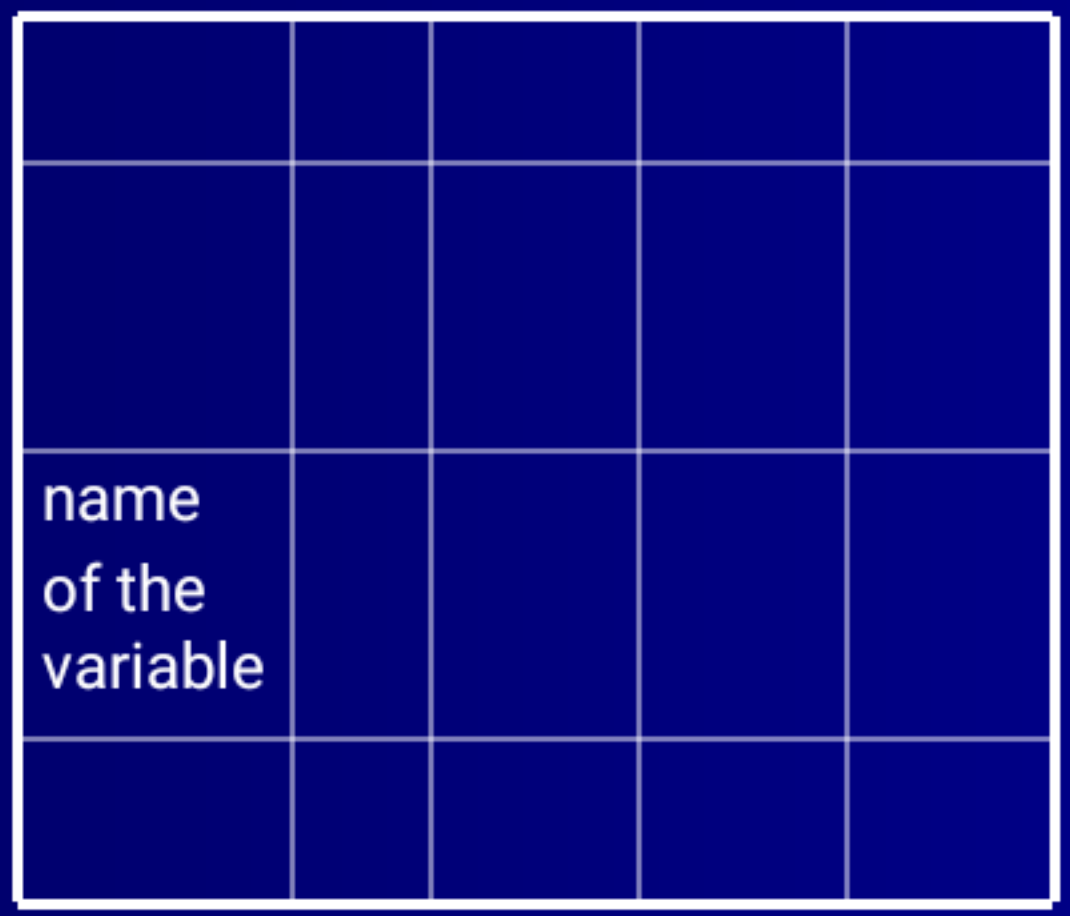
Variable

Variable



Variable

- Pic of the memory
- 25
- 10323



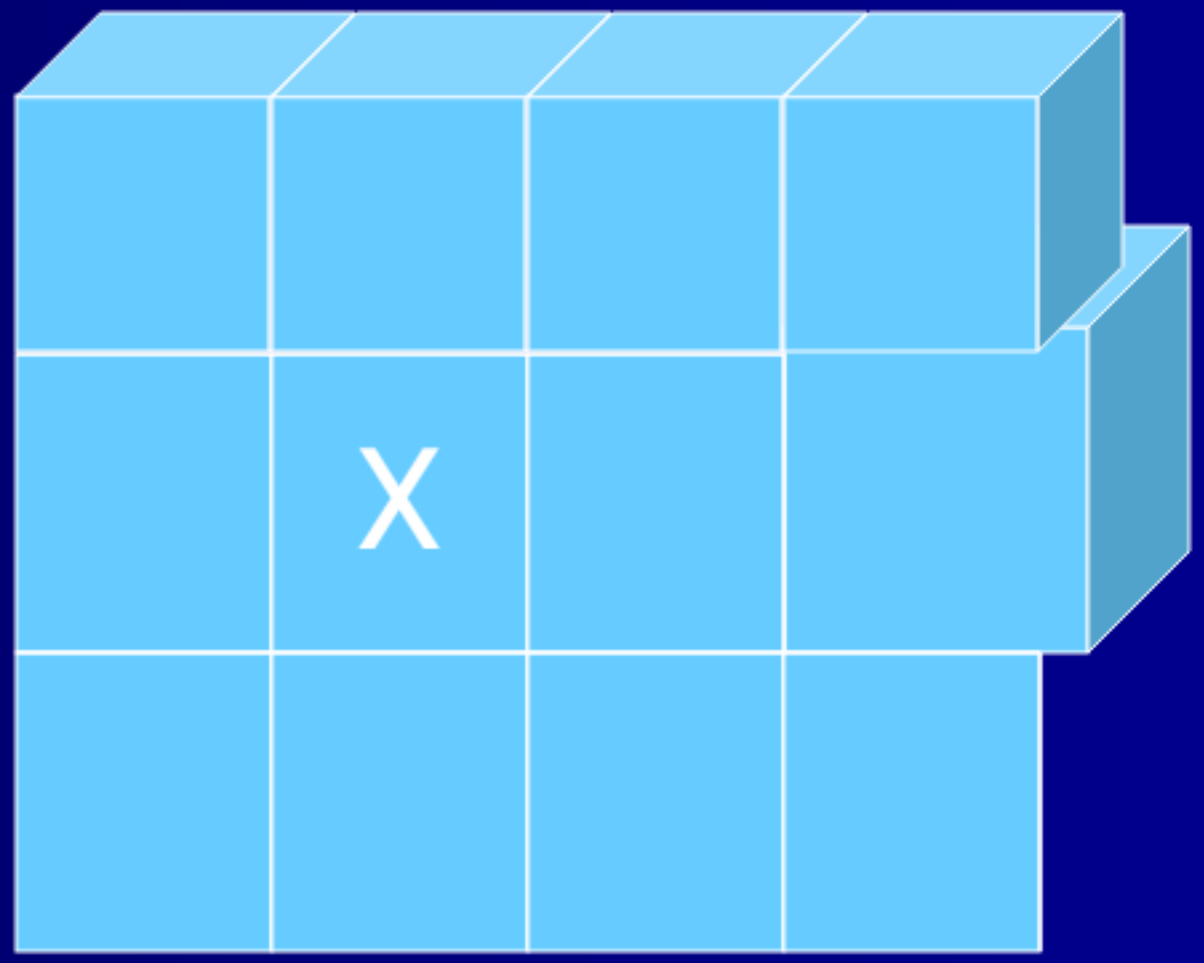
Variable

Variable starts with

- Character
- Underscore _ (Not Recommended)

Variable

- Small post box



Variable

Variable is the name of a location in the memory

e.g. x= 2;

Variable

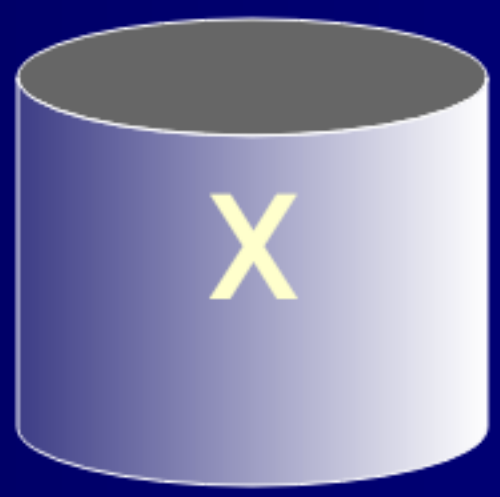
In a program a variable has:

- Name
- Type
- Size
- Value

Assignment Operator

=

x = 2



2

Assignment Operator

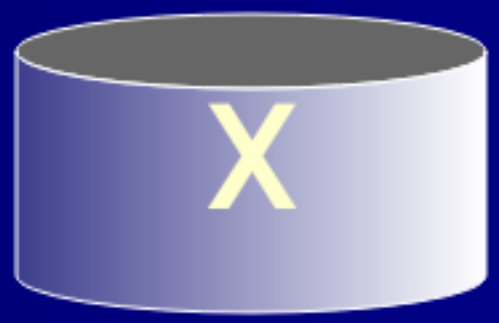
L.H.S = R.H.S.

X+ 3 = y + 4 Wrong

Z = x +4

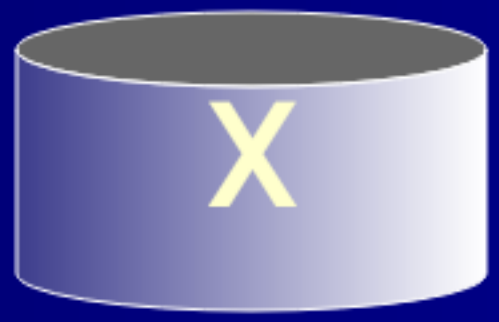
x +4 = Z Wrong

X = 10 ;



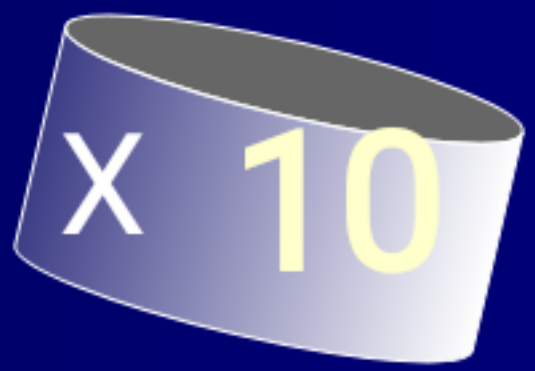
10

X = 30 ;



30

X = X + 1;

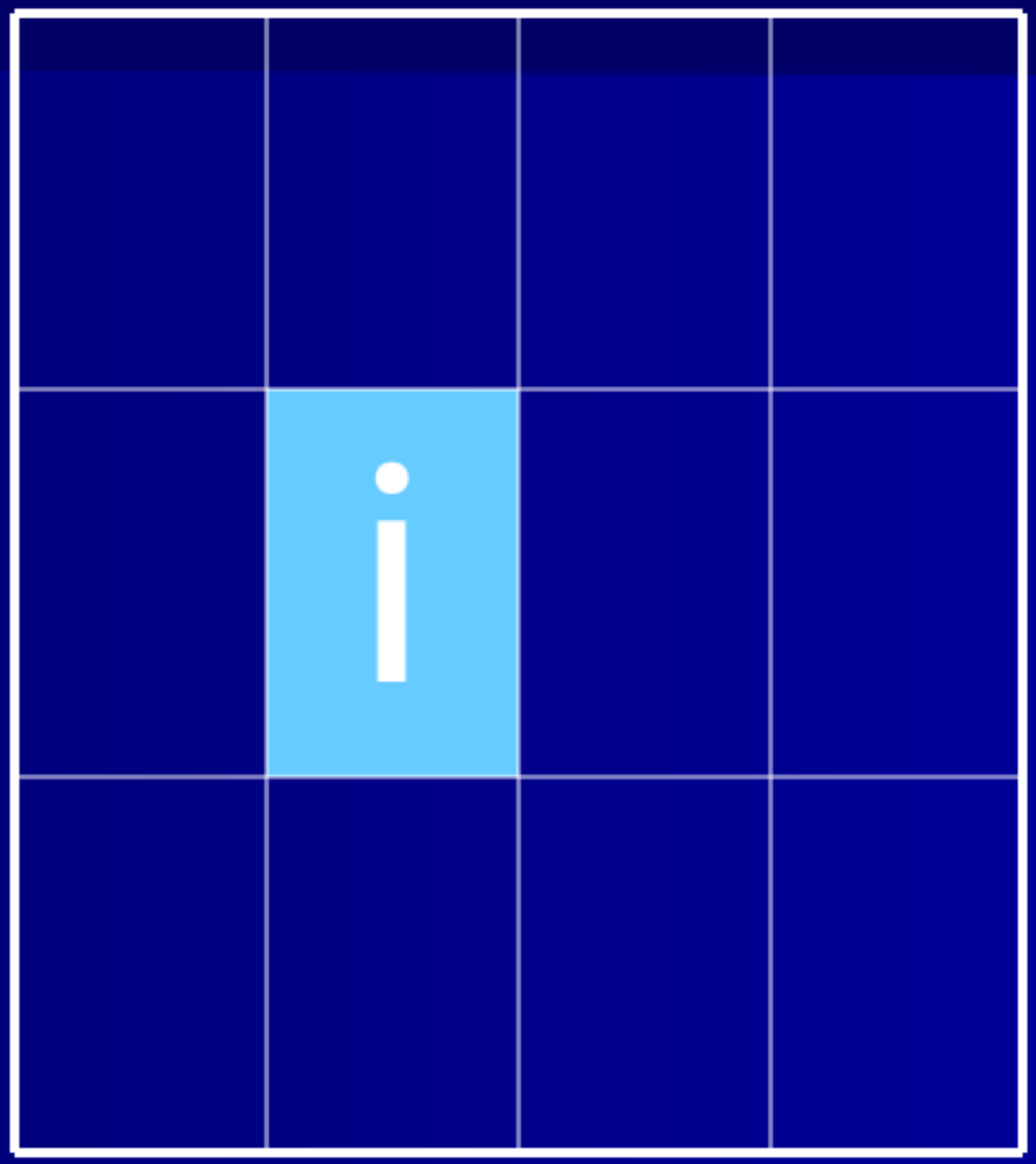


+ 1

= 11

Data type

- int i ; -> Declaration line



```
#include <iostream.h>
main ( )
{
 int x ;
 int y ;
 int z ;
 x = 10 ;
 y = 20 ;
 z = x + y ;

 cout << " x = " ;
 cout << x ;

 cout << " y = " ;
 cout << y ;

 cout << " z =x + y = " ;
 cout << z ;
}
```

int x, y, z ;

int x; int y; int z ;

Data Types

- int
- short
- long
- float
- double
- char

Arithmetic operators

Plus +
Minus -
Multiply *

Divide /
Modulus %

Arithmetic operators

```
i + j
x * y
a / b
a % b
```

% = Remainder

5 % 2 = 1

2 % 2 = 0

$$4 / 2 = 2$$

$$5 / 2 = ?$$

Precedence

- Highest: ( )
- Next: * , / , %
- Lowest: + , -